

Connected component analysis based two zone approach for Bangla character segmentation

Tasnim Zahan*
tithy.cse@gmail.com

Mohammad Reza Selim*
selim@sust.edu

Mohammad Shahidur Rahman*
rahmanms.bd@gmail.com

Muhammed Zafar Iqbal*
mzi@sust.edu

**Department of Computer Science and
Engineering
Shahjalal University of Science and
Technology
Sylhet, Bangladesh*

Abstract— In this paper a complete and robust segmentation process for printed Bangla script is presented. This phase is important for language like Bangla in which resources are not easily available. In Bangla OCR system, one of the main challenges is character segmentation. Proper segmentation increases the accuracy of recognition process. Segmentation is difficult as there are more than 300 basic, modified and compound characters in the script and they are topologically connected. Some characters are overlapped. We have used connected component analysis based two zone approach that can addresses all the problems associated with character segmentation of Bangla script with less effort. In our system we have tried to identify the matra line properly. Then using this information we have divided the script into two zones (upper and lower zone). Connected components from each part are then collected by overlooking matra line. Using some heuristics segmented character parts are organized for proper recognition. Proposed method solves the problems of character segmentation in Bangla script and thereby increases the accuracy of segmentation phase that leads to a greater accuracy of overall optical character recognition (OCR) system.

Keywords— *OCR, Segmentation, Bangla Language Processing, connected component analysis*

I. INTRODUCTION

In the field of OCR research, a perfect segmentation is required before individual characters are recognized. As Bangla is a low-resource but inflectional language we are therefore interested in an approach that will minimize effort, but still produce systems that are acceptable to users. In segmentation, basic constituent of a script, which are certainly characters, are extracted from an image. Number of characters in Bangla is large (more than 300) and two or more characters combine to form new character shapes called compound character. Along with, modifiers makes it more difficult to segment in the character level. Some characters or part of the characters lied on the upper part of the matra line, some are at the lower part, below the base line. Most of them are overlapping characters. So it is really challenging task to segment Bangla script in the character level.

Many researchers have addressed the issue of character segmentation of Bangla script. Various methods have been proposed by different authors.

Chaudhuri & Pal [1] proposed the first complete Bangla OCR system for printed documents. Using horizontal and vertical projection profile analysis and headline removal technique they segmented the documents into lines, words, and characters. They also divided the words into three zones using headline and base line detection techniques.

Mahmud et al [2] proposed a recognition process for isolated as well as continuous printed Bangla characters. This process also supports multi font Bangla script recognition. Preprocessing involves segmentation at various levels. They used piecewise linear scanning, greedy search algorithm and base line detection technique for character segmentation.

Mahmud et al [3] used projection profile based method for segmenting Bangla script. For character segmentation they used vertical scanning method. Depth First Search (DFS) algorithm also used for overlapped character segmentation.

Mehedi et al [4] proposed a new approach using characteristic functions and hamming network for segmentation and recognition of printed Bangla text. They used a set of characteristic functions for segmenting upper portion (above matra line) and characters that come under the Base line (three zone approach). It also uses a combination of Flood-fill and Boundary-fill algorithm for segmenting some characters that cannot be segmented using traditional approach.

Sattar et al [5] proposed a condition based approach for overlapped character segmentation. They analyze the width of a single character, if it is big enough then checked for the overlapped character ekar ‘০’. They don’t talked about other kind of overlapping.

Rownak et al [6] used curved scanning for overlapped character segmentation.

Garain & Chaudhuri [11] proposed a statistical analysis based method for segmenting touching characters in printed Bangla script. They noted that touching characters occur mostly at the middle of the middle zone and by inspecting the pixel patterns and their relative position with respect to the predicted middle zone certain suspected points of touching were found. The geometric shape is cut at these points and the OCR scores are noted. The best score gives the desired result.

In most the above mentioned work, they divided the words into three zones, but error in base line detection reduces the overall accuracy of the segmentation process. There are a lot of issues related with overlapped character segmentation and unfortunately none of the talked methods resolve these issues completely.

In this paper we tried to address all the problems associated with character segmentation. We used simple but effective techniques for solving these issues. In the proposed method we have used two zone approach for reducing complexity and connected component analysis technique applied for solving the problems related to overlapped character segmentation.

II. SOME PROPERTIES OF BANGLA SCRIPT

In Bangla script there are 11 vowels and 39 consonant (total of 50) characters, which are called basic characters. The concept of upper/lower case is absent in Bangla. Characters often take modified shapes which are called modifiers or allographs. Both vowel and consonant modifiers are present in the script. Another thing that create problem for Bangla OCR is compound character. Several consonant characters may combine to form compound characters that partly retain the shape of the constituent characters [6]. Most compound characters are formed by joining two consonants although three or more consonants can also be combined. There are more than 250 compound characters. Because of them, total number of characters (basic, modified and compound) to be recognized in this script increases to about 300. In Bangla, there exist a horizontal line at the upper part of the script (for most of the characters) is called the 'matra' line or head line. In a word characters are connected through this line which makes character segmentation of Bangla script very difficult. In order to do this, detection of matra line is one of the major task in segmentation. Some example of Bangla characters:

Compound Character	Formation of the Character
ন্ড	ন+ড
ক্ট	ক+ট
ভ্র	ভ+র
ঞ	ষ+ট+র

- Some characters with matra: ক, ড, ঘ
- Some characters without matra: ঞ, ঞ, গ, ণ

III. PROPOSED METHOD

A typical OCR system pipeline consists of reprocessing, segmentation, recognition and post-processing parts. In this work we focused on the segmentation process which is the back bone of the recognition phase.

A. Preprocessing

Text digitization is done by a flatbed scanner having resolution 100 to 300 dpi. At first the image is converted into grayscale and using bilateral filters noises are removed. Then through binarization image is converted to two-tone image. Image binarization converts an image of up to 256 gray levels to a black and white image. Here we have used Otsu's threshold technique.

When a document page is fed to the scanner, it may be skewed by a few degrees. For skew correction we used the technique described in [7]. In this method skew is corrected using stroke-whitespace zoning mechanism. Horizontal projection method is the fundamental actuation of this method where projection of the document is considered in every angle. The image is first cropped and all the white spaces outside the minimum fit rectangle are eliminated. The document is then scanned from top to bottom to find the angle that contains maximum number of white lines (between text lines) and then rotated accordingly. Skew correction process is shown in fig. 1.

To solve the problem of 180° rotated images, matra line feature of Bangla is being used. This line divides the words into two zones. In normal case, upper zone < lower zone. But in such case when upper zone > lower zone, then the image need to be rotated by 180°.

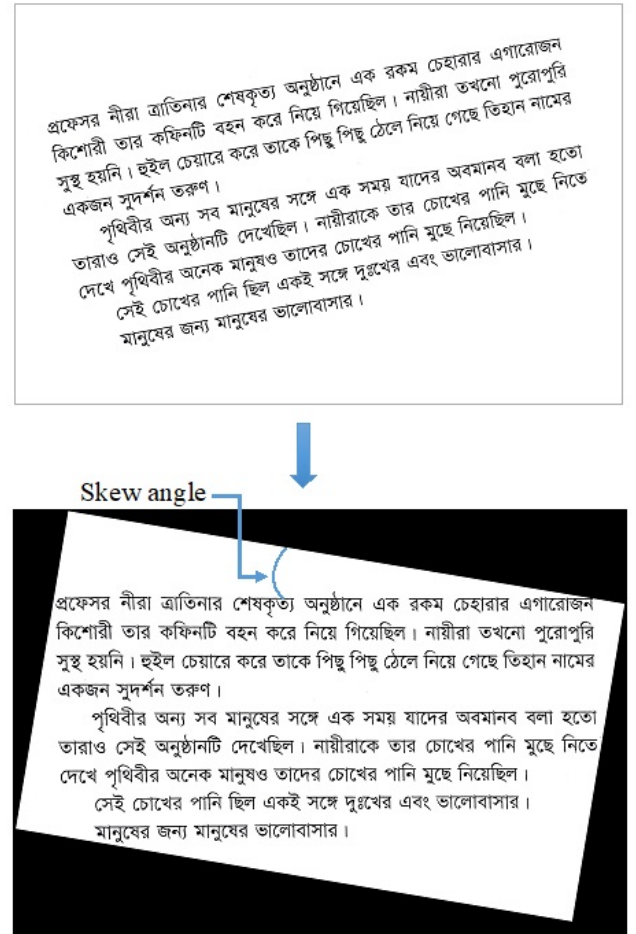


Figure 1: skewed Bangla printed text (at the top) and corrected skew (at the bottom)

B. Line segmentation

In Bangla script, text lines are almost of the same height. There is a narrow horizontal gap between two lines. These gaps are easily detected by horizontal projection profile (HPP), number of black pixels in each row. Text lines are then easily separated by detecting the valleys in HPP. Line segmentation process is shown in fig. 2.

প্রফেসর নীরা ত্রাতিনার
কিশোরী তার কফিনটি
সুস্থ হয়নি। হুইল চেয়া
একজন সুদর্শন তরুণ

Figure 2: Line segmentation process

In some cases lower part of one line (like ‘০’, ‘২’, ‘৩’, lower part of ‘ছ’, ‘হ’, etc.) overlapped with the upper part of next line (upper part of ‘ই’, ‘ট’, ‘ৈ’, etc.). To handle this, we have used the statistical approach. First average line height, *mode*, is calculated. Then comparing this value with each line, overlapped lines are spotted and separated in their middle zone with lowest HPP and threshold value.

C. Word segmentation

Vertical projection profile (VPP) is used in word segmentation from extracted lines. In this process each line is scanned vertically and number of black pixels in each column is calculated. In vertical scan if no black pixel found then that is considered as the gap between words. Word segmentation process is shown in fig. 3.

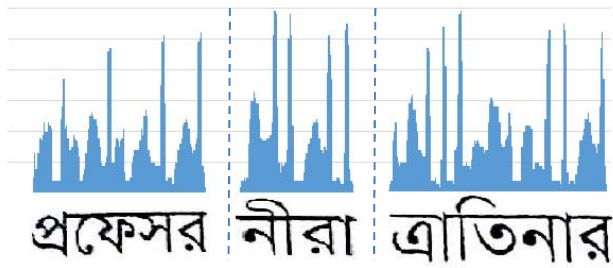


Figure 3: Word Segmentation

D. Character segmentation

Character segmentation is the most difficult part of overall segmentation process. Two major characteristics of Bangla script which make it more difficult and erroneous in case of character level segmentation are – *Matra line* and *Character overlap*. Bangla words are connected through *matra (head) line*. In our approach, based on the *matra line* words are divided into two zones – upper and lower, shown in fig. 4.

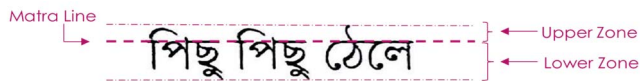


Figure 4: Zone separation

Upper zone denotes the portion above the *matra line* which may contain parts of basic and compound characters, parts of vowel and consonant modifiers, nasalization sign, apostrophe sign. Lower zone covers the part below the *matra line*. By removing *matra line*, characters get topologically disconnected [fig. 5].

পুরোপুরি

Figure 5: Disconnected Characters

Then using the connected component method, individual components of a word are separated [fig. 6]. Here we have used chain approximation algorithm with 8-connectivity. Connected component analysis method also overcome the problem associated with overlapped characters.

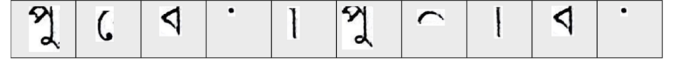


Figure 6: Segmented Characters (sorted based on horizontal position)

E. Matra line detection

Overall accuracy of segmentation process is very much dependent on the proper detection of *matra line* as it is the core component of character segmentation. If we take the horizontal projection profile (used in line separation) of a line or a word, then the highest number of black pixel denotes the *matra line*. But it may take more than one row. For that we have used the method described in [8]. In this method first order difference of horizontal projection profile (HPP) is used for detecting *matra line* region. By analyzing the HPP of different lines of text, it is noticed that number of black pixels in *matra line* region is much higher than the rows just above and below that region. So the first order difference clearly shows the two peak points (positive and negative) which denotes the starting and ending points of the *matra line* region respectively.

For example, the HPP values of the upper half of a text line:

[0, 0, 0, 0, 0, 0, 2, 4, 9, 88, 92, 41, 32, 35, 38, 46, 52]

Their first order differences are:

[0, 0, 0, 0, 0, 2, 2, 5, 79, 4, -51, -9, 3, 3, 8, 6]

Row 9 is the starting point of the *matra line* and row 11 is the end. Graphs in fig. 7 and 8 shows the peak points.

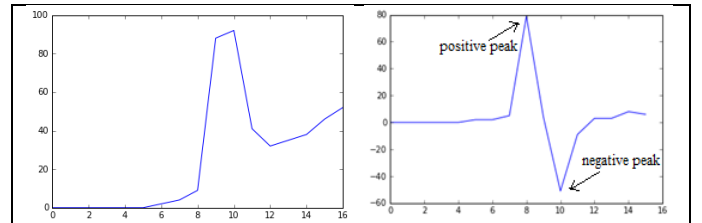


Figure 7: HPP histogram

Figure 8: First order difference

F. Problem analysis of character segmentation

Matra line removal can create some problems. In this process characters like g(গ), N(ণ), sh(শ), e(এ), O(ও), kh(খ), j(জ), th(থ) can be split into two sub-segments [fig. 9]. To join

the sub-segments we consider that all characters in the lower zone touches the base line. If the left sub-segment does not touch the base line, the algorithm combines it with the right sub-segment (character like ঞ, ণ, ঞ etc.). Again if a segment does not touch the base line and it is fully overlapped with another segment (left or right) which touches the base line then they are combined together (character like ঞ, ঞ, ঞ etc.).

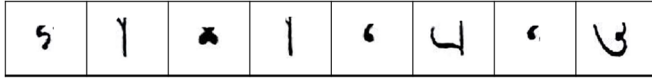


Figure 9: Characters splits into two parts

IV. RESULT ANALYSIS

We have tested our system with variety of scanned images from different books. Our line segmentation and word segmentation process provides 100% accuracy. Character segmentation provides 98.33% accuracy due to some touched and broken characters. Line, word and character segmentation results based on 3 books ('Muktijuddher Itihas' by Md. Zafar Iqbal, 'Kakababu O Aschorjo Dip' by Sunil Gangopadhyay and 'Gabbu' by Md. Zafar Iqbal) are shown in table 1, 2 and 3 respectively.

Table 1: Line segmentation result

SN.	Number of lines	Line segmentation accuracy (%)
Book 1	655	100
Book 2	2610	100
Book 3	348	100

Table 2: Word segmentation result

SN.	Number of words	Word segmentation accuracy (%)
Book 1	6472	100
Book 2	22018	100
Book 3	3237	100

Table 3: Character segmentation result

SN.	Number of Segments	Character segmentation accuracy (%)
Book 1	35909	99
Book 2	110,251	97
Book 3	15973	99

V. CONCLUSION

In this paper we proposed a combination of simple but robust techniques that can solve almost all kinds of problem associated with segmentation of Bangla Script. Our novel two zone approach reduces the complexity as well as increased the accuracy. Connected component analysis reduces the overall error rate caused by overlapped characters. We have considered all most 300 characters and our target is to recognize all these characters properly using deep machine learning technique.

REFERENCES

- [1] B. B. Chaudhuri and U. Pal, "A Complete Printed Bangla OCR System," Pattern Recognition, Vol. 31, No. 5, pp. 531–549, 1998.
- [2] J. U. Mahmud, M. F. Raihan and C. M. Rahman, "A complete OCR system for continuous Bengali characters," In: Proceedings of the TENCON, 1372–1376, 2003.
- [3] S.M. Milky Mahmud, Nazib Shahrier, A.S.M Delowar Hossain, Md. Tareque Mohmud Chowdhury, and Md. Abdus Sattar, "An Efficient Segmentation Scheme for the Recognition of Printed Bangla characters," Proceedings of ICCIT, pp 283-286, 2003.
- [4] Md. Al Mehedi Hasan, Md. Abdul Alim, and Md. Wahedul Islam "A New Approach to Bangla Text Extraction and Recognition from Textual Image", Proceedings of ICCIT, 2005.
- [5] Md. Abdus Sattar, Khaled Mahmud, Humayun Arafat and A. F. M. Noor Uz Zaman, "Segmenting Bangla Text for Optical Recognition," ICCIT, 2007. ISBN: 978-1-4244-1550-2, pp. 1-6.
- [6] Ahnaf Farhan Rownak, Md. Fazle Rabby, Sabir Ismail, and Md. Saiful Islam, "An Efficient Way for Segmentation of Bangla Characters in Printed Document Using Curved Scanning" 5th International Conference on Informatics, Electronics and Vision (ICIEV), 2016.
- [7] Mohammad Abu Obaida, Tanay Kumar Roy, Md. Abu Horaira and Md. Jakir Hossain, "Skew Correction Function of OCR: Stroke-Whitespace based Algorithmic Approach," International Journal of Computer Applications 28(8):7-12, August 2011.
- [8] Subhra Sundar Goswami, "Identification of Matra Region and Overlapping Characters for OCR of Printed Bengali Scripts," Intelligent Computing and Information Science, Volume 135 of the series Communications in Computer and Information Science, 2011, pp. 606-612.
- [9] Shamim Ahmed and Mohammad Abul Kashem, "Enhancing the Character Segmentation Accuracy of Bangla OCR using BPNN," International Journal of Science and Research (IJSR), ISSN (Online): 2319-7064, Volume 2 Issue 12, December 2013.
- [10] Md. Abul Hasnat, S. M. Murtoza Habib, Mumit Khan, "A High Performance Domain Specific OCR for Bangla Script," Novel Algorithms and Techniques in Telecommunications, Automation and Industrial Electronics, pp. 174-178, Springer Science+Business Media B.V. 2008.
- [11] U. Garain and B. B. Chaudhuri, "Segmentation of touching characters in printed Devnagari and Bangla scripts using fuzzy multifactorial analysis," IEEE Trans. Syst. Man Cybern., Part C 32: 449–459, 2002.